Enhanced Keypoint-Based Approach for Identifying Copy-Move Forgery in Digital Images

Girish Singh Bisht
Department of Computer Science & Engineering
Tula's Institute
Dehradun , India 248001
singhgiri2037@gmail.com

Arpit Jain
Department of Computer Science & Engineering
Koneru Lakshamaiah Education Foundation
Vadessahwaram, A.P., India
dr.jainarpit@gmail.com

Vandna bansla
Shivalik Engineering College
Dehradun , India 248001
vbansala1@gmail.com

Khemraj Sharma
Department of Business Administration, Doctoral Program in Management Sciences
Nanhua University, Dalin,
Chiayi, 62248, Taiwan (R.O.C.)
khemraj8517@gmail.com

Rudramani Bhutia
Department of Computer Science & Engineering
Koneru Lakshmaiah Education Foundation,
A.P., India
rudra.kuna@gmail.com

Vikas Kumar
School Engineering and Computing
Dev Bhoomi Uttrakhand University
Dehradun, Uttrakhand, India
saini.vikaskumar86@gmail.com

2024 7th International Conference on Contemporary Computing and Informatics (IC3I) | 979-8-3503-5006-7/24/$31.00 ©2024 IEEE | DOI: 10.1109/IC3I61595.2024.10829331

***Abstract* - Copy-move forgeries often exploit homogeneous regions in images with large-scale attacks to either highlight or conceal target objects. These manipulations are simple to execute but challenging to notice. Forgery detection techniques like Copy-Move Forgery Detection (CMFD) cannot detect these forged documents because they are unable to identify a sufficient number of effective keypoints in homogeneous areas, leading to inaccurate and inefficient results. SURF stands for Speeded-Up Robust Features and is used in this paper along with A-KAZE and Scale-Invariant Feature Transforms. According to our experiment, A-KAZE offers superior detection performance under diverse attacks, especially when it comes to large-scale attacks targeting homogeneous regions. A-KAZE is found to be more accurate than SIFT, SURF, and A-KAZE when applied to the NB-CASIA dataset, achieving detection accuracies of 89.2%, 93.9% and 98.98%.)**

Keywords— Data Analysis, Forgery Detection, Image Processing, Image attack detection

I. INTRODUCTION

With today's sophisticated image editing software, faking digital images has never been easier. Data sharing has become increasingly important thanks to digital data, especially images. The accessibility of budget-friendly digital cameras has simplified image capture. The impact of social networking platforms is substantial, with individuals frequently sharing and forwarding images. It turns every user into a potential source of digital content.

Furthermore, the prevalence of user-friendly image editing software empowers both beginners and experts to craft plausible modifications. These alterations might range from harmless modifications for amusement to more insidious image manipulations aimed at deception [1]. This reality introduces doubt into the credibility of digital images. Especially in the case of news dissemination, research publications, and legal proceedings, these digital images become critical. In the last few years, digital data that can be viewed as images has played a significant role in providing information. Image capture became easier with the availability of low-cost digital camera devices. People's lives have been affected significantly by social networks, as evidenced by statistics. Nowadays, people upload their photos online and forward others' photos[2]. Digital information is likely to be available to every user because of this.

It is convenient for novices and experts to use simplistic image-editing Software, which allows them to make authentic alterations. Among these tweaks, some are made for entertainment, while others are done maliciously [3], [4]. Digital images are uncertain because of this fact. These digital images become even more problematic when they are used for specific purposes such as news broadcasts, research journals, and courtrooms.

Forgeries involving copy-move are also known as copy-paste forgeries or cloning forgeries. They involve copying and pasting segments within an image [5], [6]. It is common for forgers to conceal or emphasize an element in an image. In order to create convincingly forged documents, image-processing operations like JPEG compression, noise addition, rotation, and scaling may be used during copying and moving. Due to the ease with which copy-move operations can now be conducted, Software for image editing has become increasingly important and active with CMFD [7], [8]. The two most commonly used methods for detecting copy-moves are block-based detection and interest keypoint detection [9], [10].

Forgeries using copy-move techniques involve copying and pasting specific parts of an image into another. Due to the fact that clones can exist anywhere, it is impossible to find all forged parts. Furthermore, the copy area is from the same image with compatible features (such as color and noise). Due to this, detecting differences in features is difficult. Therefore, the detection of copy-move forgeries remains challenging [11], [7] (CMFD). CMFD has two general objectives:

Copy-move forgeries have been detected through a variety of methods, most of which are block-based [12], [13], [14], A deep learning-based method, and a feature point-based method [15], [16], [17]. A geometric and post-processing attack operation makes it impossible to detect copy-move regions using block-based algorithms. In many other fields, deep learning-based methods have proven successful, but not in CMFD, where extensive training data is required.

979-8-3503-5006-7/24/$31.00 ©2024 IEEE

Authorized licensed use limited to: REVA UNIVERSITY. Downloaded on October 09,2026 at 07:02:45 UTC from IEEE Xplore. Restrictions apply.

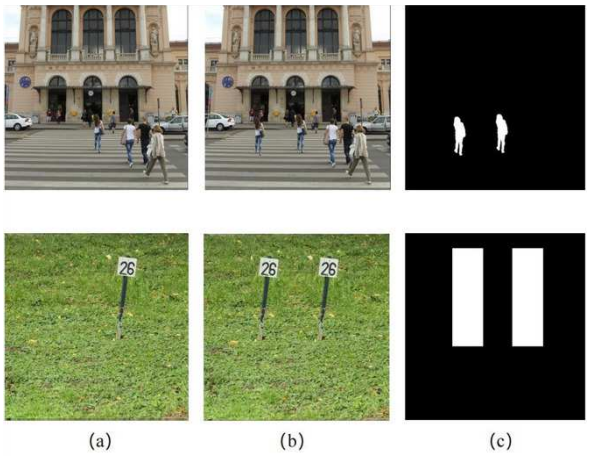


Fig. 1. From left to right: original image (a), copy–move forged image (b), and binary ground truth overlying copy–move area (c)

II. Literature Survey

A keypoint set is detected, described, and matched for tampering detection. Keypoint sets are described based on a variety of features. SIFT (Scale Invariant Features Transform) eliminates geometric distortion, illumination distortions, etc [15]. Further, [9] presented a Jlinkage algorithm that employs a robust clustering of geometry transformations in order to enhance the model [18]. In [19], it is proposed that speeded robust features (SURF) be used as the basis for a CMFD scheme, and a K-D tree be used to identify similar SURF descriptors [20]. There is no explicit indication of the boundaries of the duplicated regions, but lines indicate the duplicated regions. When it comes to creating feature sets for large images, keypoint-based methods are much more efficient. As a result, the keypoint-based method builds and matches the FV based on the keypoint sets that have been detected. There is a considerable improvement in the speed of SURF feature detection. Their speed is two to three times faster than SIFT's. BFs and integral images are the cause of this. As soon as identical keypoint descriptors are identified, BF is used to remove mismatches based on predetermined thresholding parameters.

In the subsequent study, [21] a robust clustering phase was introduced to the CMFD approach. In [22], for CMFD, It was demonstrated how to integrate both image blocks and interest points in an adaptive over-segmentation approach. According to [23], SIFT-symmetry, symmetry-based matching, and SIFT-based CMFD are combined to create a robust CMFD method. [21] describes the process of detecting maximally stable color regions and representing them as Zernike moments. The multiplicity of the same file is then handled using a modified matching algorithm that takes into account n best match features. In [24], three forensic detectors were devised based on the anomalies in keypoint distribution, revealing if keypoints had been removed globally or locally. We then apply the detectors to a practical image forensics scenario where we need to detect copy-move forgeries using SIFT-based methods. In [25], Harris corners were detected by a two-stage detection method based on ordered gradient histograms and hue histogram descriptors extracted from multi-support regions. In [26], As a result of the combination of KAZE with SIFT, an increased number of interest points is generated.

In order to cluster similar regions on images, over-segmentation is performed on them. Labelled keypoints are then processed to extract SURF and PCT features. According to [27], SURF features and PCET can be used to identify keypoints in images and detect regions with tapering. Geometrically transformed images can be detected, but larger-scale changes are more challenging. For improved detection accuracy, several operations are applied to image keypoints. Using keypoints and segmentation [28] addressed images with a variety of textures. PCET and SURF keypoint are used together. The method cannot detect forged images with scale; it also requires much time to perform some steps. As part of the SIFT feature, sensitivity-based clustering is also integrated [29]. There are two approaches used in [30], ELA and colour filter arrays, as well as modified and unmodified images, which are required for both methods. Incorporating two different analytical methods, they are able to detect both copy-moves and splices. In [31-34], Two steps were involved in calculating image features: first, using the SIFT keypoint detector, and then using the Fourier-Miller transform. In the case of fewer in-lines than a threshold, an FMT algorithm is applied to the entire image.

III. Proposed Methodology

In our paper, we propose detecting copy-move forgeries using keypoints. In addition to reducing false positives, it improves the complexity of time and space.

A. SURF

There is no need to worry about lighting changes, affine transformations, scaling, or rotation of the SURF. We use Hessian matrix determinants to determine SURF's scale and location. A scaled representation of the Hessian matrix $H(x;\sigma)$ can be seen in Equation (1).

$$H(x;\sigma) = \begin{bmatrix} C_{xx}(x,\sigma) & C_{xy}(x,\sigma) \\ C_{xy}(x,\sigma) & C_{yy}(x,\sigma) \end{bmatrix} \quad (1)$$

It is not uncommon to treat $C_{xx}(x,\sigma)$ as a similarity measure for convolutions of two Gaussian second-order derivatives: $\partial^2 g(\sigma)/\partial x^2$ and $C_{xx}(x,\sigma)$.

In order to reduce computation time, the box filter simulates Gaussian second-order derivatives, $D_{xx}(x,\sigma)$, $D_{xy}(x,\sigma)$, and $D_{yy}(x,\sigma)$, which improves the calculation speed of convolution and reduces its complexity. Equation (2) shows how to approximate Hessian's determinant:

$$\det(H_{approx}) = D_{xx}D_{yy} - (0.9D_{xx})^2 \quad (2)$$

In the construction of image pyramid scales, integral and box filters are used. With a neighbourhood of interest points, we determine non-maximum suppression $3X3X3$ interest points. Using our method, we eliminate candidate points with an approximate Hessian matrix smaller than T_{SURF}, which measures how fast Hessian responses are.

Before obtaining rotation-invariant descriptors, it is necessary to determine the main direction of rotation. A response centring at a diameter of 6 meters determines the pointed direction. In order to obtain the Haar wavelet response, the circular region is filtered using a Haar wavelet filter. Wavelet responses centred at a point with an angle of 1/2 are drawn as fan-shaped areas. Using Haar wavelets, fan-shaped regions can be calculated. This vector is selected as the main direction among all the others.

Authorized licensed use limited to: REVA UNIVERSITY. Downloaded on October 09,2026 at 07:02:45 UTC from IEEE Xplore. Restrictions apply.

B. Accelerated-KAZE (AKAZE

CMF detection using the AKAZE algorithm can be used to detect key points within uniform areas, especially when detecting key points within uniform areas. In this procedure, the image's data is adaptively blurred using nonlinear scale space, so noise is reduced without interrupting the image's data. When non-linear diffusion is used, luminance is modelled by increasing the scale levels. Flow functions are diverging in a partially differential equation that controls diffusion. A nonlinear partial differential equation with nonlinear scale-space is responsible for the diffusion of luminance in an image. Here is an equation for nonlinear diffusion:

CMF detection using the AKAZE algorithm can be used to detect key points within uniform areas, especially when detecting key points within uniform areas. In this procedure, the image's data is adaptively blurred using nonlinear scale space, so noise is reduced without interrupting the image's data. When non-linear diffusion is used, luminance is modelled by increasing the scale levels. Flow functions are diverging in a partially differential equation that controls diffusion. A nonlinear partial differential equation with nonlinear scale-space is responsible for the diffusion of luminance in an image. Here is an equation for nonlinear diffusion:

$$\frac{\partial LI}{\partial t} = div(C(x,y,t)\nabla LI \tag{3}$$

Scale parameter t signifies the image scale, gradient operator is gradient, divergence is divergence, luminance is LI, and conductivity is $C(x,y,t$, which is given by

$$C(xyt) = G(|\nabla LI_{\sigma}(xyt)|) \tag{4}$$

A smooth Gaussian version of an image is called LI_{σ}, and its gradient is called LI_{σ}. Our study evaluates one of Perona and Malik's two conductivity function. Using AKAZE as a reference, G_2 conductivity functions that support larger areas compared to smaller ones:

$$G_2 = \frac{1}{1|\nabla LI_{\sigma}|^2\Omega^2} \tag{5}$$

In relation to edge information, Ω controls the diffusion level. In general, the lower the value, the more edge information will be retained. The pattern is derived empirically by taking 70% of the gradient histogram as the Ω. Throughout the AKAZE, scale images are generated using fast explicit diffusion (FED). Simple explicit techniques like FED make use of fluctuating time step sizes, which makes them extremely faster than common explicit techniques. In order to improve rotation invariance, AKAZE detectors detect radiation using the Scharr filter and Hessian matrix, and we have the determinant. As a result, AKAZE's descriptor relies on MLDB, which is a highly effective algorithm. In this research, AKAZE was used because it is invariant to scale, rotation and distinctive at different scales.

C. Scale-invariant feature transform (SIFT)

The Analyzing a 128-dimensional feature vector can identify SIFT features. Following are the four steps that must be completed in order for the process to be completed:

1) Scale-space extrema detection:

Gaussian Differences (DoG) are first used to detect scale space extremes. This image is defined in terms of its scale space $L(W,V,\sigma)$, which is the convolution of the Gaussian function $Gf(W,V,\sigma)$ with the input image $Y(W,Y)$.

$$L(W,V,\sigma) = Gf(W,V,\sigma) * Y(W,V) \tag{6}$$

With

$$Gf(W,V,\sigma) = \frac{1}{2\pi\sigma^2} e^{-(W^2+V^2)/2\sigma^2} \tag{7}$$

This image is composed of W and V coordinates. $Y(W,V), \sigma$ represents the standard deviation of the Gaussian function $Gf(W,V,\sigma)$.

Differential Gaussian (DoG) techniques are used to determine the image's key points based on the equation given:

$$\begin{aligned} D(W,V,\sigma) &= \big(Gf(W,V,k\sigma) - Gf(W,V,\sigma)\big) * Y(W,V) \\ &= L(W,V,\sigma) - L(W,V,k\sigma) \\ &\quad - L(W,V,\sigma) \end{aligned} \tag{8}$$

An initial image $Y(W,V)$ is convoluted with a Gaussian blur $Gf(W,V,k\sigma)$ (at $k\sigma$.) to form $L(W,V,\sigma)$. For $D(W,V,\sigma)$ local maxima and minima, pixels are compared with eight neighbour pixels at the same scale as well as nine neighbour pixels up and down. Keypoint candidates are pixels with the lowest or highest value among all pixels.

2) Keypoint localization:

It tends to generate a lot of candidate keypoints, some of which are unstable, as a result of scale-space extreme detection. Consequently, after identifying key points by comparing pixels with their neighbours, they are interpolated for increased precision. To obtain more accurate key point locations, Taylor series expansions of the second order are used. In the following example, we expand $D(W,V,\sigma)$ as a Taylor expansion, assuming a candidate keypoint as the origin:

$$D(w) = D + \frac{\partial D^T}{\partial w} w + \frac{1}{2} w^T \frac{\partial^2 D^T}{\partial w^2} w \quad - \tag{9}$$

Using W as a candidate keypoint, D and its derivatives can be calculated, while $w = (W,V,\sigma)$ Represents a shift from this keypoint. In order to determine the extremum $\widehat{w}$, we can take the derivative forward and set it to zero. (see Figure 1):

$$\widehat{w} = -\frac{\partial^2 D^{-1}}{\partial w^2} \frac{\partial D}{\partial w} \tag{10}$$

When $\widehat{w}$ is higher in any dimension, compared to previous keypoints, the extremum lies closer to a different keypoint. A new keypoint will be chosen for the candidate, and interpolation will be carried out around it. An extremum position can be estimated by adding the final offset $\widehat{w}$ to the candidate keypoint's position. By estimating DW as a second-order Taylor expansion, it is estimated that key points with low contrast will be removed at the offset. Hence, low-contrast points should be removed. Excluding key points occurs when the intensity of an extremum falls below a certain threshold. As a result, the localization of keypoint candidates is accomplished by eradicating low-contrast contrast keypoints during the process.

3) Orientation assignment:

Lastly, we assign orientations based on local gradients in the image. A local image property assigns orientation information for each key point. A Gaussian image L must first be calculated. Next, we calculate the magnitude gradient m and orientation n:

Authorized licensed use limited to: REVA UNIVERSITY. Downloaded on October 09,2026 at 07:02:45 UTC from IEEE Xplore. Restrictions apply.

$$m(W,V) = \sqrt{(L(W+1,V)-L(W-1,V))^2+(L(W,V+1)-L(W,V-1))^2} \quad (11)$$

$$\theta(W,V) = \tan^{-1}\left(\frac{L(W,V+1)-L(W,V-1)}{L(W+1,V)-L(W-1,V)}\right) \quad (12)$$

Gradient orientations of sample points generate orientation histograms, and the peak locations in these histograms determine further orientation.

Keypoint descriptor: The last step is the creation of keypoint descriptors by using the local gradient data that was used above. The key point is located in a 16 by 16 neighbourhood that is divided into 16 subblocks of 4 by 4 each. Orientation histograms are generated for each sub-block. Consequently, 128 bin values can be accessed. Descriptors for keypoints are formed as vectors. The robustness of the system is also enhanced through several measures such as noise addition, rotation, and scaling.

IV. Results and Analysis

After We present our results, analyze them, and compare them to those of existing experiments in this section. Simulations are performed using Python image processing library version 3.9 and OpenCV 4.8.1.78's image processing library. Windows 11 Home is installed on the laptop, which is powered by an Intel Core i3 processor running at 2.19GHz. In this experiment, we evaluated our algorithm on the dataset NB-CASIA. CMFD experiments are highly dependent on parameters. To achieve optimum outputs, we used a variety of parameters and thresholds. Keypoint threshold parameters are 0.0005 for AKAZE feature detectors in this method.

In FIGURE 2-4, This method detects both single-copy-move forgeries as well as multiple-copy-move forgeries. The output image with the detected tampered area can also be seen in Figures 2-3, which is based on the natural dataset. Detecting the tampered area in a smooth region is another challenging task when detecting copy-move forgeries. An image is processed by the AKAZE, SURF, and SIFT algorithms. As far as visual results are concerned, we can determine smooth regions fairly well using our method. The default parameters of A-KAZE, SIFT, SURF, and KAZE do not allow feature extraction in tampered regions. These points can, however, be reached by SURF and a-KAZE at small thresholds. In order to use CMFD with keypoints, the number of points obtained must be sufficient.

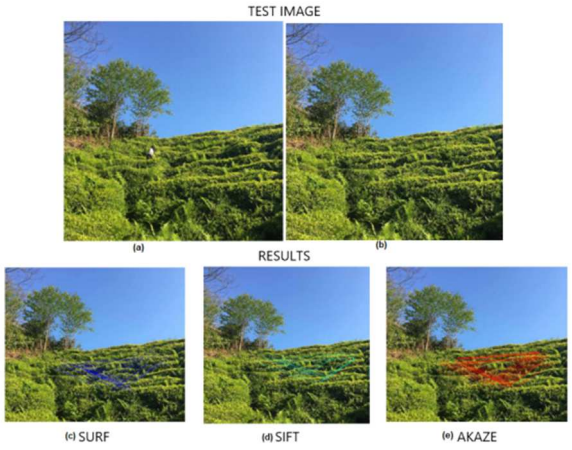


Fig. 2. Results of mountain hills: (a) unique image; (b) meddled image; (c) SURF detector; (d) SIFT detector; (e) AKAZE detector



Fig. 3. Results of cats images: (a) unique image; (b) meddled image; (c) SURF detector; (d) SIFT detector; (e) AKAZE detector.

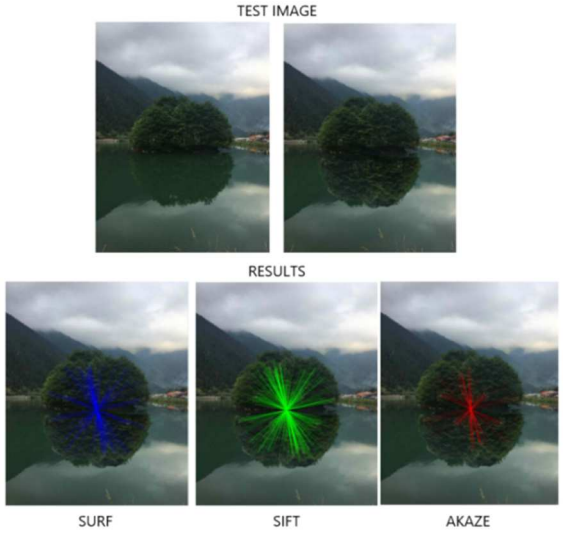


Fig. 4. Results of rotated-tree-lake images: (a) unique image; (b) meddled image; (c) SURF detector; (d) SIFT detector; (e) AKAZE detector.

CMFD approaches based on keypoints were compared experimentally to the proposed method. It was more accurate than existing method recalled more data, and was more accurate than others. A multi-copies-moves test of the proposed method is conducted as well.

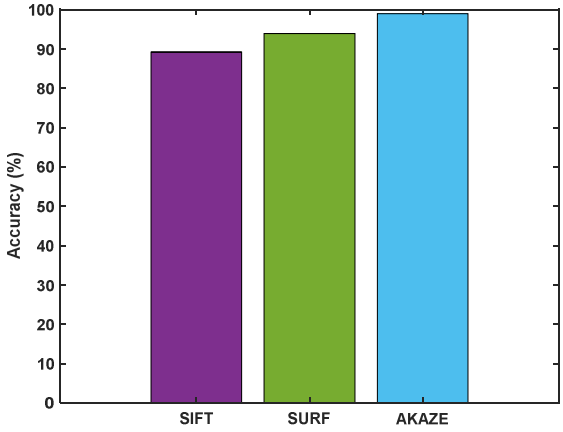


Fig. 5. Accuracy (%) of SIFT, SURF and AKAZE.

Authorized licensed use limited to: REVA UNIVERSITY. Downloaded on October 09,2026 at 07:02:45 UTC from IEEE Xplore. Restrictions apply.

V. CONCLUSION

Changing the location of an image on the screen conceals tampering without being obvious. These types of attacks can tamper with images, which makes forensics concerned about their authenticity. Image manipulation has become easier over the past few years thanks to powerful software development. The use of forgery detection techniques has been the subject of many research papers. It is also impossible to detect smooth forged regions with keypoint-based methods when there are insufficient keypoints. The forgery of copy-moves is discussed in this work. We evaluated three keypoint-based approaches: SIFT SURF and A-KAZE. Our experiments demonstrated that the A-KAZE scheme significantly outperforms the others, providing superior detection performance under various attack scenarios. The A-KAZE approach excels particularly in detecting forgeries within homogeneous regions affected by large-scale attacks. When applied to the NB-CASIA dataset, A-KAZE achieved the highest detection accuracy at 98.98%, compared to 89.2% with SIFT and 93.9% with SURF, while also maintaining the lowest computational cost. A-KAZE has the potential to offer substantial improvements over traditional methods when it comes to copy-move forgery detection, as demonstrated by these results. This advancement can enhance the reliability and effectiveness of CMFD tools in various practical applications, ensuring better integrity and authenticity of digital images.

REFERENCES

[1] H. Farid, "Digital forensics in a post-truth age," Forensic Sci. Int., vol. 289, pp. 268–269, 2018.

[2] P. Rani, P. N. Singh, S. Verma, N. Ali, P. K. Shukla, and M. Alhassan, "An implementation of modified blowfish technique with honey bee behavior optimization for load balancing in cloud system environment," Wirel. Commun. Mob. Comput., vol. 2022, pp. 1–14, 2022.

[3] H. Ding, L. Chen, Q. Tao, Z. Fu, L. Dong, and X. Cui, "DCU-Net: a dual-channel U-shaped network for image splicing forgery detection," Neural Comput. Appl., vol. 35, no. 7, pp. 5015–5031, Mar. 2023, doi: 10.1007/s00521-021-06329-4.

[4] P. Rani, N. Hussain, R. A. H. Khan, Y. Sharma, and P. K. Shukla, "Vehicular intelligence system: time-based vehicle next location prediction in software-defined internet of vehicles (SDN-IOV) for the smart cities," Intell. Things AI-IoT Based Crit.-Appl. Innov., pp. 35–54, 2021.

[5] K. Sitara and B. M. Mehtre, "Digital video tampering detection: An overview of passive techniques," Digit. Investig., vol. 18, pp. 8–22, 2016.

[6] S. Verma et al., "An automated face mask detection system using transfer learning based neural network to preventing viral infection," Expert Syst., p. e13507, 2024.

[7] Christlein, C. Riess, J. Jordan, C. Riess, and E. Angelopoulou, "An evaluation of popular copy-move forgery detection approaches," IEEE Trans. Inf. Forensics Secur., vol. 7, no. 6, pp. 1841–1854, 2012.

[8] M. A. Qureshi and M. Deriche, "A bibliography of pixel-based blind image forgery detection techniques," Signal Process. Image Commun., vol. 39, pp. 46–74, 2015.

[9] N. B. Abd Warif et al., "Copy-move forgery detection: survey, challenges and future directions," J. Netw. Comput. Appl., vol. 75, pp. 259–278, 2016.

[10] H. Li, W. Luo, X. Qiu, and J. Huang, "Image Forgery Localization via Integrating Tampering Possibility Maps," IEEE Trans. Inf. Forensics Secur., vol. 12, no. 5, pp. 1240–1252, May 2017, doi: 10.1109/TIFS.2017.2656823.

[11] R. Dixit and R. Naskar, "Review, analysis and parameterization of techniques for copy–move forgery detection in digital images," IET Image Process., vol. 11, no. 9, pp. 746–759, Sep. 2017, doi: 10.1049/iet-ipr.2016.0322.

[12] J. Ouyang, Y. Liu, and M. Liao, "Robust copy-move forgery detection method using pyramid model and Zernike moments," Multimed. Tools Appl., vol. 78, no. 8, pp. 10207–10225, Apr. 2019, doi: 10.1007/s11042-018-6605-1.

[13] N. B. Abd. Warif, Mohd. Y. I. Idris, A. W. Abdul Wahab, R. Salleh, and A. Ismail, "CMF-iteMS: An automatic threshold selection for detection of copy-move forgery," Forensic Sci. Int., vol. 295, pp. 83–99, Feb. 2019, doi: 10.1016/j.forsciint.2018.12.004.

[14] M. Emam, Q. Han, and X. Niu, "PCET based copy-move forgery detection in images under geometric transforms," Multimed. Tools Appl., vol. 75, no. 18, pp. 11513–11527, Sep. 2016, doi: 10.1007/s11042-015-2872-2.

[15] Y. Rodriguez-Ortega, D. M. Ballesteros, and D. Renza, "Copy-Move Forgery Detection (CMFD) Using Deep Learning for Image and Video Forensics," J. Imaging, vol. 7, no. 3, p. 59, Mar. 2021, doi: 10.3390/jimaging7030059.

[16] X. Wang, C. Wang, L. Wang, H. Yang, and P. Niu, "Robust and effective multiple copy-move forgeries detection and localization," Pattern Anal. Appl., vol. 24, no. 3, pp. 1025–1046, Aug. 2021, doi: 10.1007/s10044-021-00968-y.

[17] J. Yang, Z. Liang, Y. Gan, and J. Zhong, "A novel copy-move forgery detection algorithm via two-stage filtering," Digit. Signal Process., vol. 113, p. 103032, Jun. 2021, doi: 10.1016/j.dsp.2021.103032.

[18] X. Wang, G. He, C. Tang, Y. Han, and S. Wang, "Keypoints-Based Image Passive Forensics Method for Copy-Move Attacks," Int. J. Pattern Recognit. Artif. Intell., vol. 30, no. 03, p. 1655008, Mar. 2016, doi: 10.1142/S0218001416550089.

[19] B. Yang, X. Sun, H. Guo, Z. Xia, and X. Chen, "A copy-move forgery detection method based on CMFD-SIFT," Multimed. Tools Appl., vol. 77, no. 1, pp. 837–855, Jan. 2018, doi: 10.1007/s11042-016-4289-y.

[20] D. Cozzolino, G. Poggi, and L. Verdoliva, "Efficient dense-field copy–move forgery detection," IEEE Trans. Inf. Forensics Secur., vol. 10, no. 11, pp. 2284–2297, 2015.

[21] C.-M. Pun, X.-C. Yuan, and X.-L. Bi, "Image forgery detection using adaptive oversegmentation and feature point matching," Ieee Trans. Inf. Forensics Secur., vol. 10, no. 8, pp. 1705–1716, 2015.

[22] N. B. A. Warif, A. W. A. Wahab, M. Y. I. Idris, R. Salleh, and F. Othman, "SIFT-symmetry: a robust detection method for copy-move forgery with reflection attack," J. Vis. Commun. Image Represent., vol. 46, pp. 219–232, 2017.

[23] Costanzo, I. Amerini, R. Caldelli, and M. Barni, "Forensic analysis of SIFT keypoint removal and injection," IEEE Trans. Inf. Forensics Secur., vol. 9, no. 9, pp. 1450–1464, 2014.

[24] L. Yu, Q. Han, and X. Niu, "Feature point-based copy-move forgery detection: covering the non-textured areas," Multimed. Tools Appl., vol. 75, no. 2, pp. 1159–1176, Jan. 2016, doi: 10.1007/s11042-014-2362-y.

[25] F. Yang, J. Li, W. Lu, and J. Weng, "Copy-move forgery detection based on hybrid features," Eng. Appl. Artif. Intell., vol. 59, pp. 73–83, 2017.

[26] Jain, A. Kumar, A. P. Shukla, H. Alshazly, H. Elmannai, A. D. Algarni, R. Kumar, and J. Yadav, "Smart communication using 2D and 3D mesh network-on-chip," Intelligent Automation & Soft Computing, vol. 34, no. 3, 2022.

[27] S. Gowroju and S. Kumar, "Robust pupil segmentation using UNET and morphological image processing," in 2021 International Mobile, Intelligent, and Ubiquitous Computing Conference (MIUCC), 2021, pp. 105-109.

[28] S. Rani, S. Gowroju, and S. Kumar, "IRIS based recognition and spoofing attacks: A review," in 2021 10th International Conference on System Modeling & Advancement in Research Trends (SMART), 2021, pp. 2-6.

[29] Jain, T. Mehrotra, A. Sisodia, S. Vishnoi, S. Upadhyay, A. Kumar, C. Verma, and Z. Illés, "An enhanced self-learning-based clustering scheme for real-time traffic data distribution in wireless networks," Heliyon, vol. 9, no. 7, 2023.

[30] T. Mehrotra and G. Sudhamsu, "Energy efficient and fault tolerant area monitoring for sensor networks," in 2023 International Conference on Distributed Computing and Electrical Circuits and Electronics (ICDCECE), 2023, pp. 1-6.

[31] Tiwari, S. Kumar, T. Mehrotra, and R. K. Singh, "Deep neural networks for comprehensive multimodal emotion recognition," in 2023 International Conference on Disruptive Technologies (ICDT), 2023, pp. 462-466.

Authorized licensed use limited to: REVA UNIVERSITY. Downloaded on October 09,2026 at 07:02:45 UTC from IEEE Xplore. Restrictions apply.

[32] G. K. Rajput, S. Kundu, T. Mehrotra, and L. Kumar, "Social contact analysis and user profiling for online shopping in Internet of Everything," in 2022 International Conference on Computational Intelligence and Sustainable Engineering Solutions (CISES), 2022, pp. 324-328.

[33] Amerini, L. Ballan, R. Caldelli, A. Del Bimbo, L. Del Tongo, and G. Serra, "Copy-move forgery detection and localization by means of robust clustering with J-Linkage," Signal Process. Image Commun., vol. 28, no. 6, pp. 659–669, Jul. 2013, doi: 10.1016/j.image.2013.03.006.

[34] P. Perona and J. Malik, "Scale-space and edge detection using anisotropic diffusion," IEEE Trans. Pattern Anal. Mach. Intell., vol. 12, no. 7, pp. 629–639, 1990.

Authorized licensed use limited to: REVA UNIVERSITY. Downloaded on October 09,2026 at 07:02:45 UTC from IEEE Xplore. Restrictions apply.